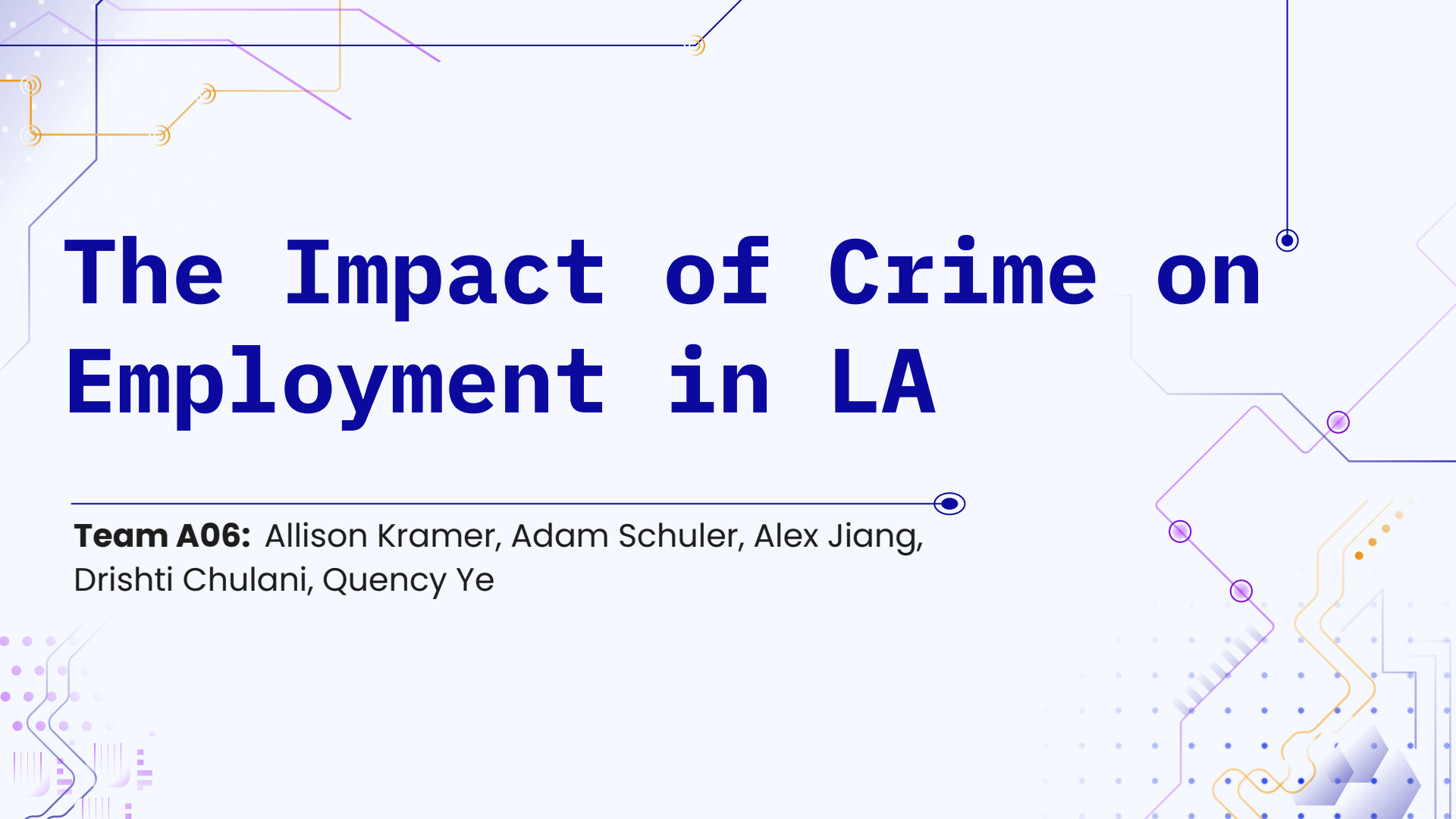




The Impact of Crime on Employment in LA

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Table of Contents

- 01** Introduction and Motivation
- 02** Problem Statement
- 03** Data Cleaning
- 04** Data Visualization
- 05** Conclusion
- 06** Limitations & Risks
- 07** Recommendations

Introduction and Problem Statement

Crime remains a major challenge in urban areas, influencing both **community safety** and **economic vitality** in Los Angeles. This project explores how employment opportunities shape these dynamics and contribute to overall **city well being**.

- Understanding how crime and employment interact helps identify patterns that guide effective policy and community action.
- We **analyzed** 15 years of LAPD crime data and 34 years of LA County Labor Trends (1990–2024).
- **Goal:** Generate insights to strengthen safety and expand opportunity across Los Angeles.



Los Angeles County: The focus area of this study

Data Overview



Crime Data (2010–2025)

Los Angeles Police Department

2 Datasets combined:

- 3M+ crime incidents
- 28 variables tracked

Key Variables:

Incident Details: DR_NO, DATE OCC, Date Rptd

Crime Classification: Crm Cd Desc, Area Name

Victim Demographics: Vict Age, Vict Sex

Location: LAT, LON

Employment Data (1990–2025)



LA County Labor Market Information

1 Dataset:

- 420 monthly records
- 5 variables

Key Variables:

Year-Month (ym)

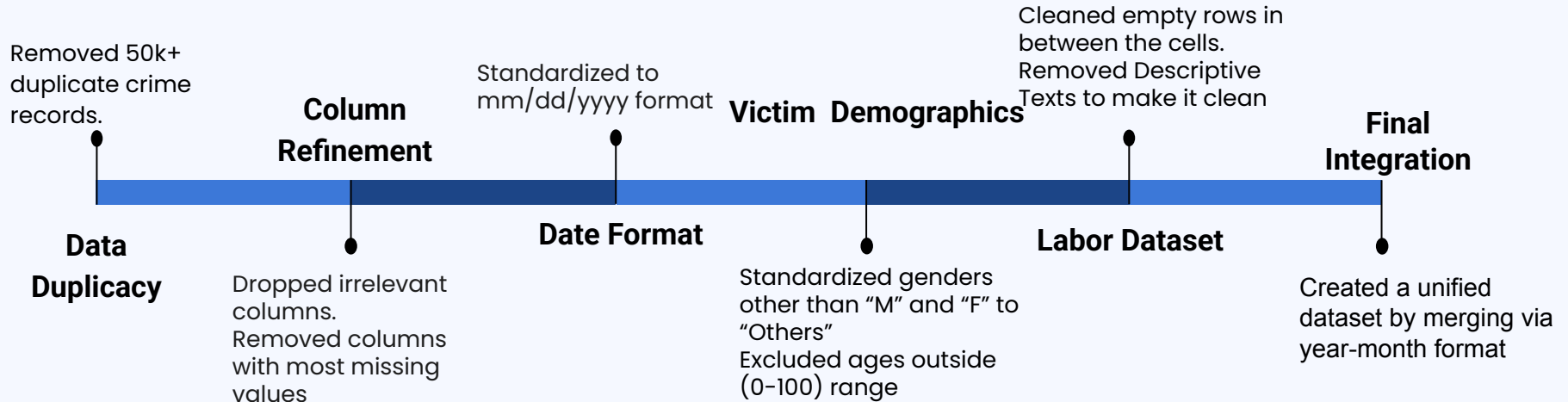
Labor Force Size

Employment count

Unemployment Size

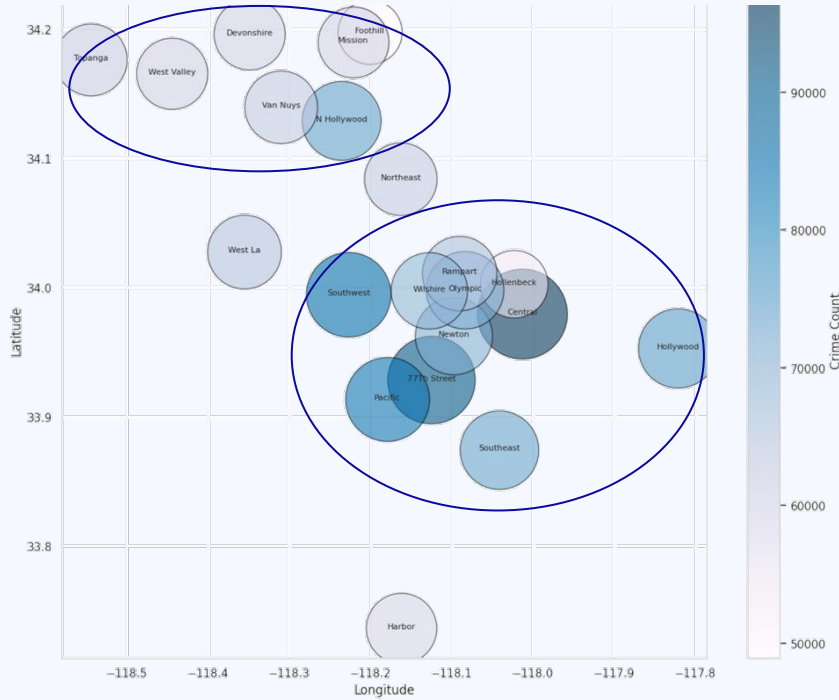
Unemployment Rate

Data Cleaning and Preprocessing



Data Visualizations

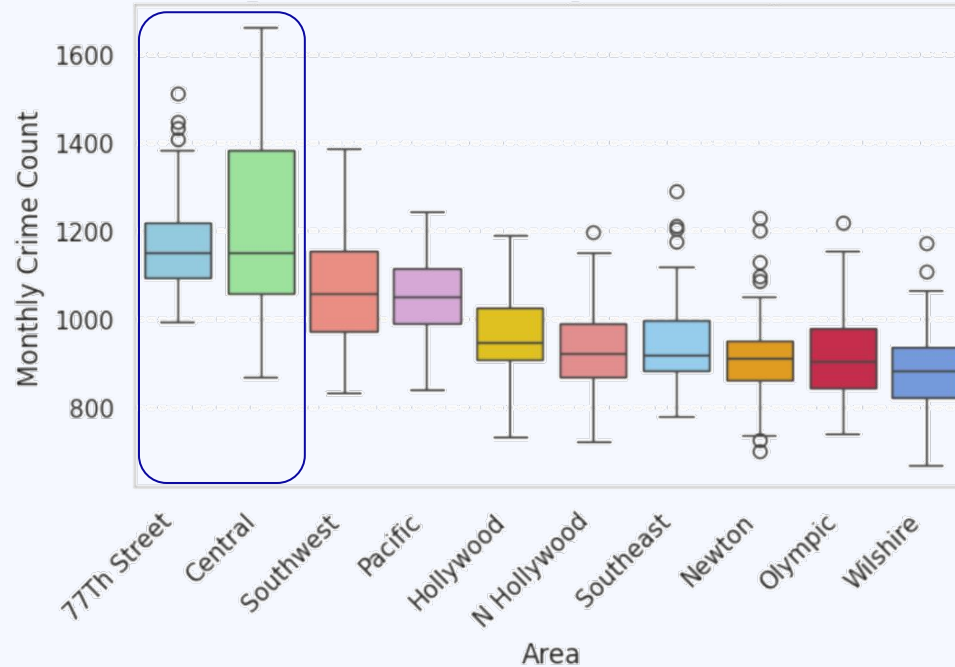
Crime Density by Area in LA (2018-2024)



- Central cluster dominance: highest crime concentration in downtown and South LA
- Distinct north-south divide: Northern areas showing lower crime than Southern and Central areas
- A core-and-surrounding pattern: crime tends to be lower in the outer areas of the city compared to the urban core

Data Visualizations

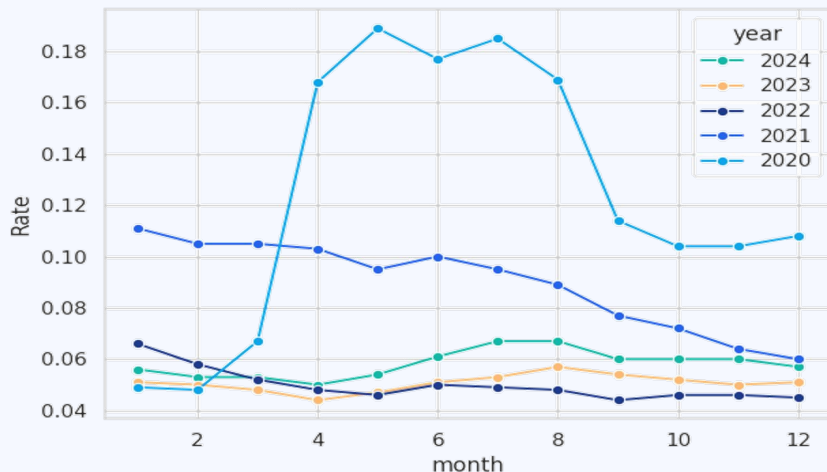
Monthly Crime Counts: Top 10 Areas (2018-2024)



- 77th Street shows consistently high monthly crime, while Central exhibits large month-to-month fluctuations with occasional extreme values
- Newton and Wilshire have shorter boxes, reflecting more stable and lower crime levels
- More rapes, more robberies, more assaults, more murders, and basement full of guns from the Article: City's Danger Zone 77th, It's Life at Mach 2

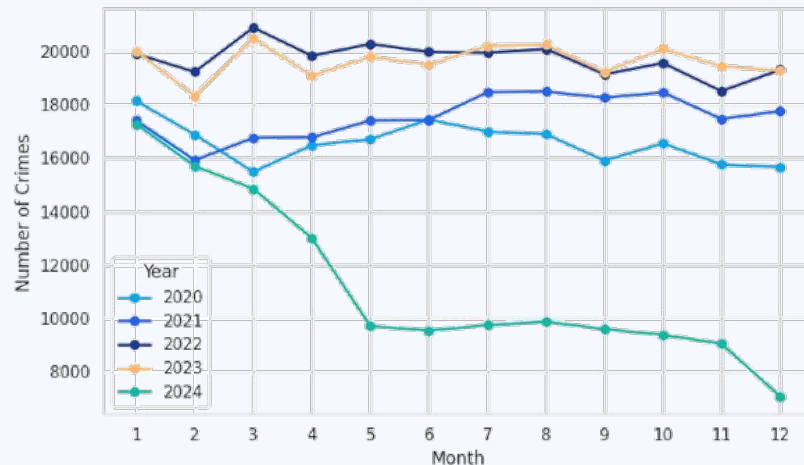
Data Visualizations

Unemployment Rate Seasonality in Recent Years



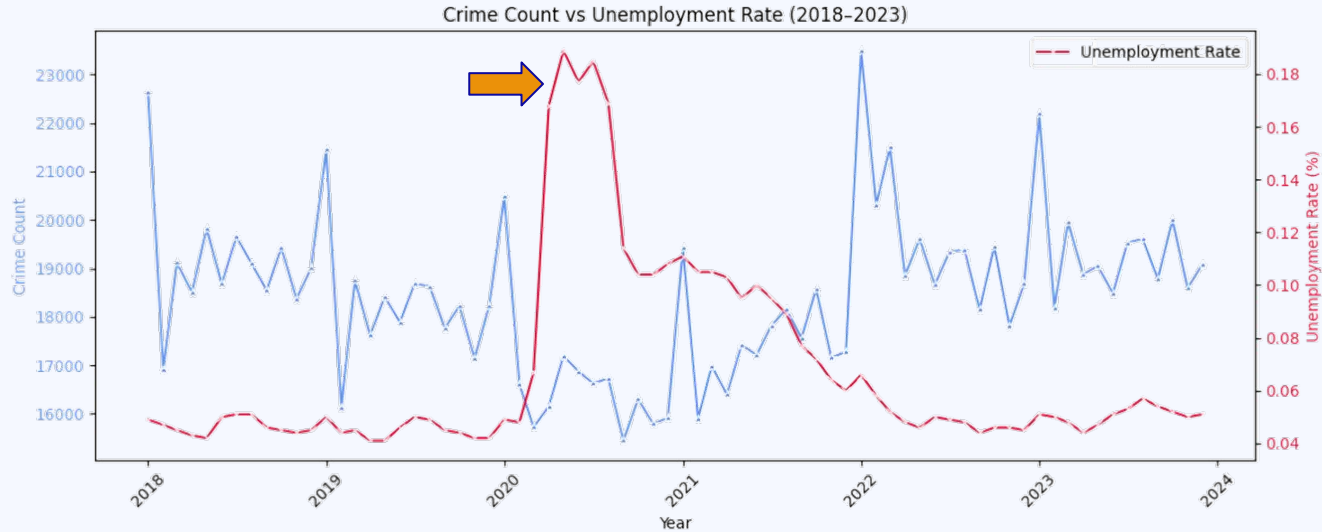
This line chart shows monthly unemployment patterns across 2020 – 2024.

Total Crimes Seasonality in Recent Years



This line chart shows monthly total crime counts across 2020 – 2024.

Data Visualizations

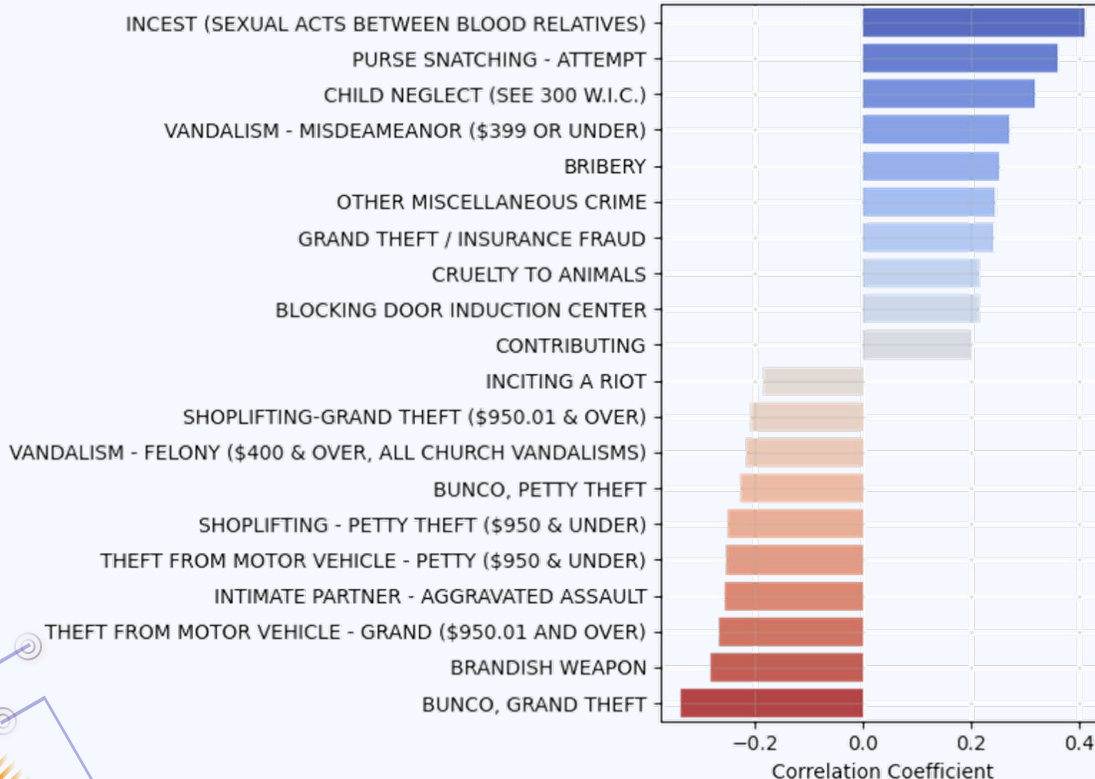


Significant **rise** in unemployment rate caused by COVID

Pre and post COVID unemployment rate remains under 6%, while crime count changes over time. Showing the both of them have **weak correlation**.

Data Visualizations

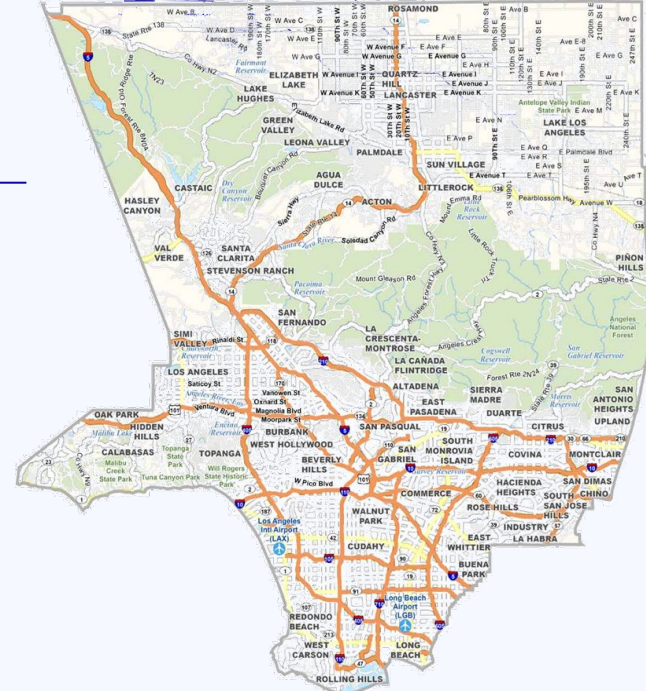
Correlation Between Unemployment Rate and Crime Types (2018-2023)



This bar chart shows the top 10 crime categories with the **strongest positive** correlations and the top 10 with the **strongest negative** correlations to the unemployment rate in Los Angeles (2018-2023).

Conclusion

1. The correlation between unemployment and crime in Los Angeles is **complex** and **weak**.
2. Crime **patterns vary** by area and offense type, with persistent hotspots regardless of employment shifts.
3. Economic downturns **influenced** both unemployment and crime, though not in a uniform way.
4. Slight seasonal fluctuations appeared in crime rates but **didn't consistently align** with economic changes.
5. The merged datasets provided descriptive insights, but **not enough evidence** to confirm a causal link.



Map of Los Angeles County: Areas Observed

Limitations & Risks

Limitation	Description
Data Quality	Missing or inconsistent records required manual cleaning.
Poor Crime Reporting	Underreporting or misclassification may skew results
Date + Time Gaps	Inconsistent formats made cross year comparisons difficult
External Factors	Policy, demographics, or economy may affect platform
Visualization	Gaps in data may distort patterns in visuals
Correlation/ Causation	No significant correlation was found; results are descriptive, not causal.

Recommendations

Expansion of Scope

Integrate other critical socio-economic data

Refine or do more in dept geographic analysis.

Expand the crime type categories

Future with Data & Analytics Focus

Improved data collection protocols

Build a real time monitoring dashboard

Develop advanced quantitative methods like forecasting methods

Policy & Strategy Alignment

Targeted Prevention Efforts

Establish cross sector governance framework

Community Education and support

Thank you !

